The safety car lottery

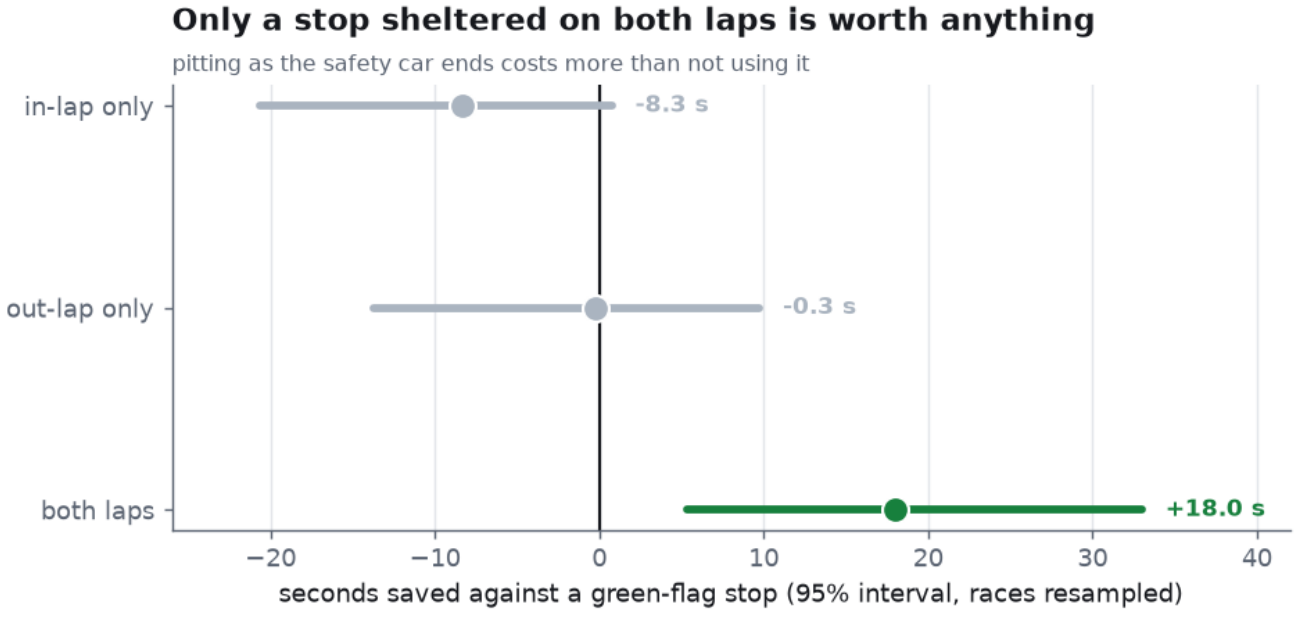
Formula 1, 2023. What a safety car is worth, who gets to use it, and how much earned advantage it deletes.

A sheltered pit stop costs 18.0 s less

A stop sheltered on both laps costs 5.2 s against 23.2 s under green — about 78% of a pit stop, handed to whoever happened to be at the right point in their stint when somebody else crashed.

23.2 s	5.2 s	+18.0 s	2,245 s
a green-flag stop	sheltered by a safety car	the prize	advantage deleted

Nothing in the source says where the safety cars were. They are found in the lap times and then told apart from the weather, which produces the same lap-time signature and the opposite effect on the order.

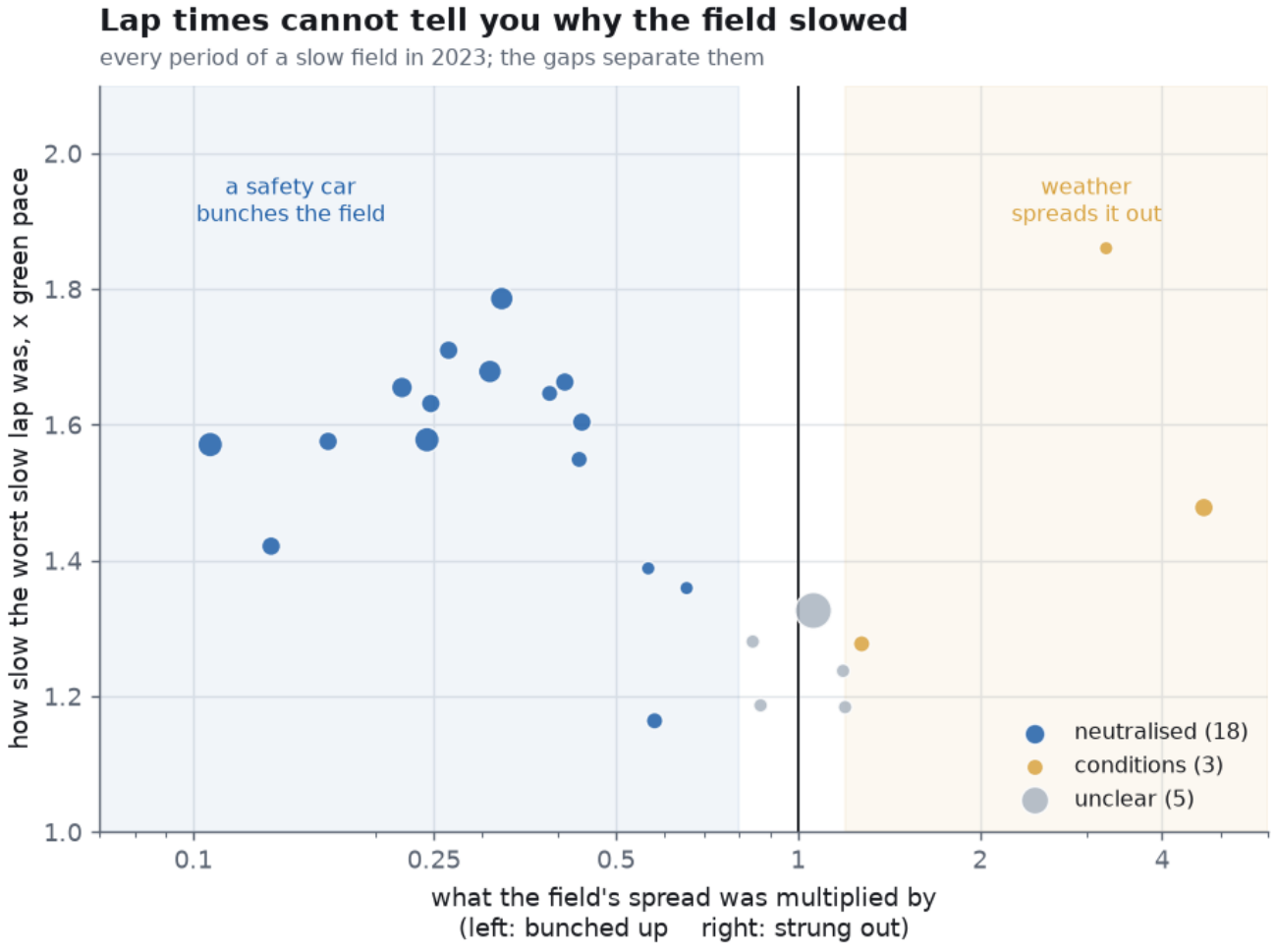


Finding them, and knowing what you found

A safety car is obvious in lap times: everybody is thirty to sixty per cent slower at once. That part is four lines of code.

The gaps separate them. Under a safety car the field is bunched up deliberately and the spread of the cars collapses, typically to a fifth. Weather does the opposite, because drivers differ far more in the wet. Same lap-time signature, opposite effect on the order.

> The hard part is that rain does the same thing, and a detector that does not ask why will price a lottery that never happened.



what	count
races	22
race-laps	1,303
laps slow	84
laps the race was stopped	6
periods	27
periods neutralised	18

Finding them, and knowing what you found (continued)

what	count
periods from conditions	3
periods unclear	5
periods still running at the flag	1

Every period the detector found

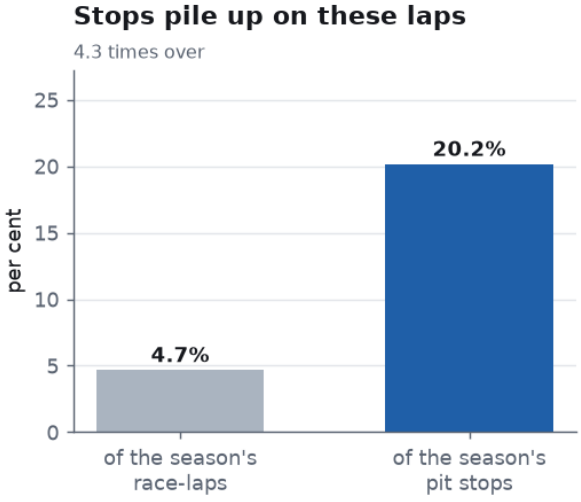
race	laps	peak	spread before	after	change	reading
r1 bahrain	41–41	1.28	56.6	47.6	0.84	unclear
r2 jeddah	18–20	1.58	31.0	5.2	0.17	neutralised
r3 albert_park	2–3	1.55	15.1	6.6	0.43	neutralised
r3 albert_park	7–9	2.33	9.2	4.8	0.53	neutralised
r3 albert_park	18–19	1.28	9.5	12.2	1.27	conditions
r3 albert_park	54–58	2.57	52.5	nan	—	to the flag
r4 baku	11–13	1.42	41.3	5.6	0.13	neutralised
r6 monaco	53–66	1.33	99.3	105.3	1.06	unclear
r8 villeneuve	8–8	1.18	5.9	7.1	1.20	unclear
r8 villeneuve	12–16	1.68	17.1	5.3	0.31	neutralised
r9 red_bull_ring	2–3	1.65	11.0	4.3	0.39	neutralised
r9 red_bull_ring	15–15	1.36	17.9	11.7	0.65	neutralised
r10 silverstone	33–38	1.57	41.2	4.4	0.11	neutralised
r13 zandvoort	2–4	1.48	8.5	39.8	4.69	conditions
r13 zandvoort	10–11	1.16	43.0	24.9	0.58	neutralised
r13 zandvoort	16–21	1.58	22.6	5.5	0.24	neutralised
r13 zandvoort	61–66	2.13	34.2	6.5	0.19	neutralised
r15 marina_bay	20–22	1.71	16.1	4.3	0.26	neutralised
r15 marina_bay	44–44	1.39	53.9	30.5	0.57	neutralised
r16 suzuka	2–4	1.66	14.9	6.1	0.41	neutralised
r16 suzuka	14–14	1.19	36.7	31.8	0.87	unclear
r17 losail	2–4	1.60	11.3	5.0	0.44	neutralised
r19 rodriguez	5–5	1.24	8.3	9.8	1.19	unclear
r19 rodriguez	33–36	1.65	27.5	6.1	0.22	neutralised
r20 interlagos	2–2	1.86	13.5	43.6	3.23	conditions
r21 vegas	2–6	1.79	20.4	6.6	0.32	neutralised
r21 vegas	26–28	1.63	20.8	5.1	0.25	neutralised

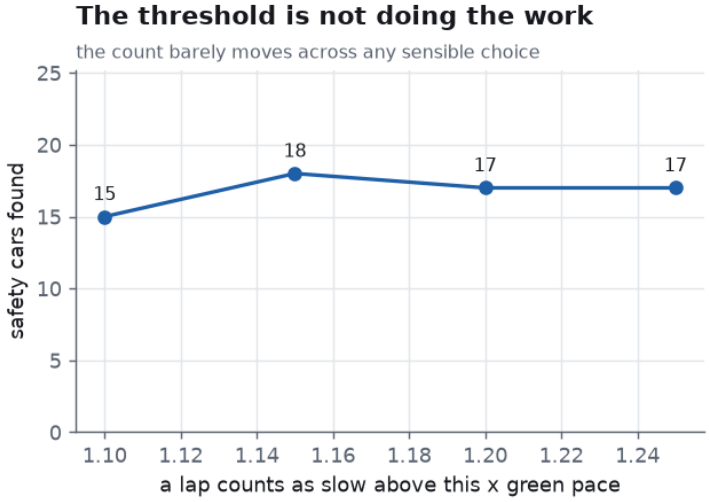
A check that uses no lap times

A safety car makes a pit stop cheap, so if these laps really are safety cars the stops should pile up on them.

20.2% of the season's 912 pit stops happened on 4.7% of its race-laps, an enrichment of 4.31 times.

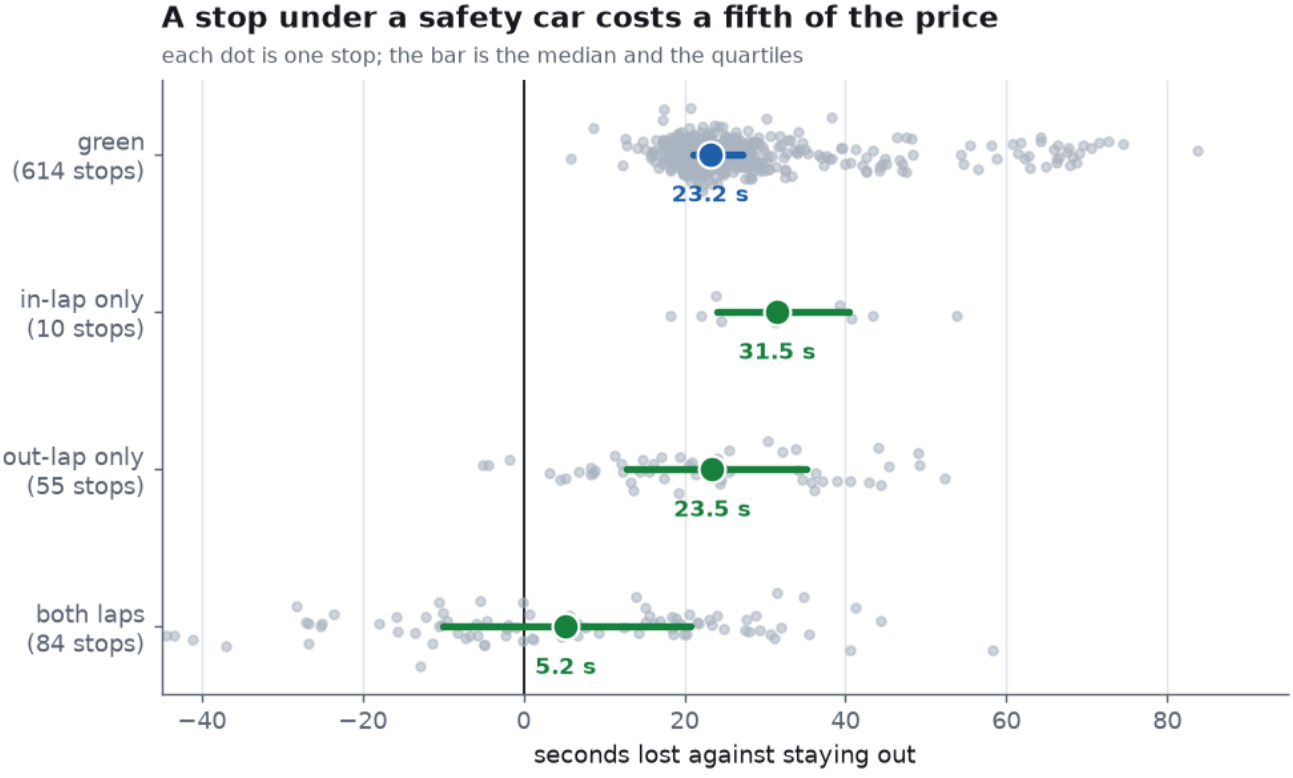
Nothing downstream depends on this number. It is here so the detector can be wrong in public.





What a stop costs

A stop with only its in-lap sheltered costs 31.5 s — more than a green one. The car dives in behind the safety car and rejoins just as the field is released, paying the pit lane's price and collecting none of the discount.



shelter	stops	median loss, s	quartiles
green	614	23.18	[21.09, 27.25]
in-lap only	10	31.49	[24.01, 40.39]
out-lap only	55	23.45	[12.73, 35.17]
both laps	84	5.21	[-10.07, 20.78]

The prize, with an interval on it

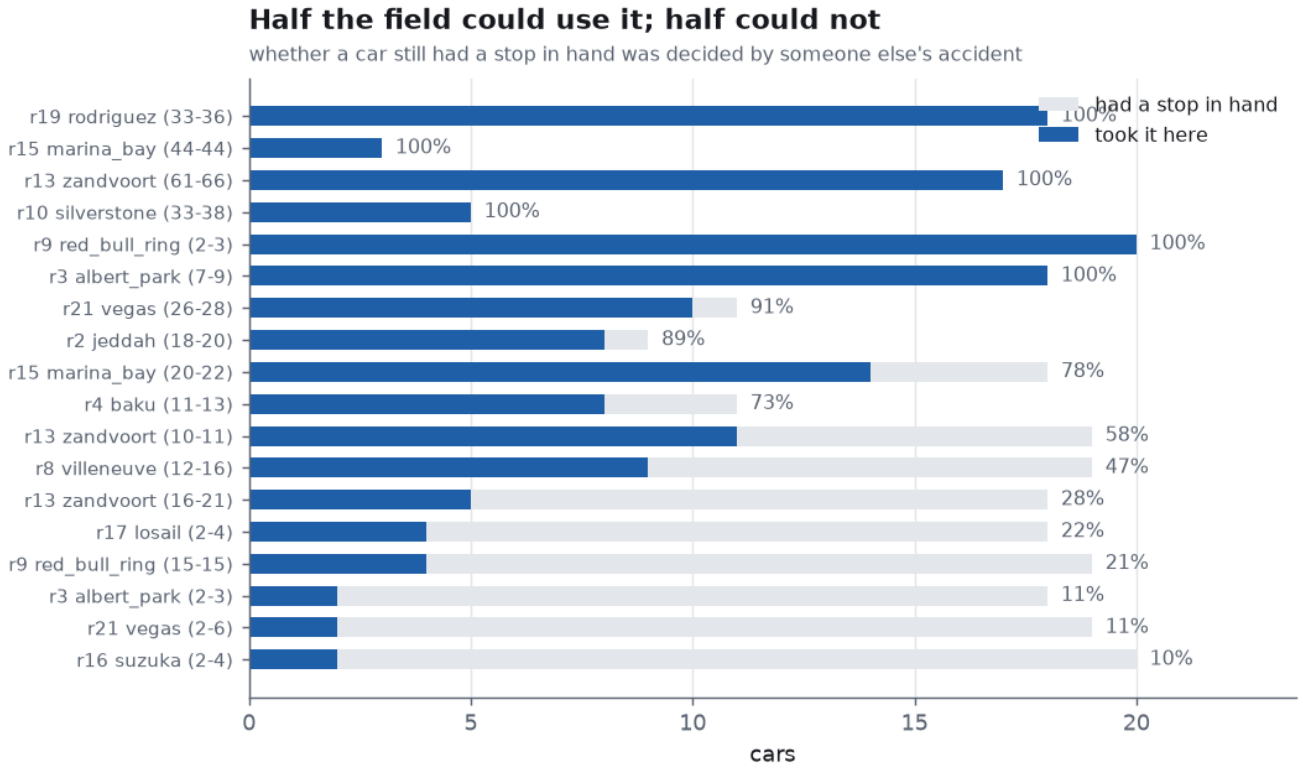
shelter	median, s	discount	95% interval	excludes zero
in-lap only	31.49	-8.31	[-20.63, +0.78]	no
out-lap only	23.45	-0.27	[-13.77, +9.75]	no
both laps	5.21	+17.97	[+5.32, +33.02]	yes

Each discount is measured against the green-flag median, and the interval is wide because eighteen neutralisations is not many.

Who was in a position to use it

Over 18 neutralisations, 160 of 280 cars that still had a stop to make took it there (57%). The rest were already on their last set and watched their rivals buy the same stop for a fifth of the price.

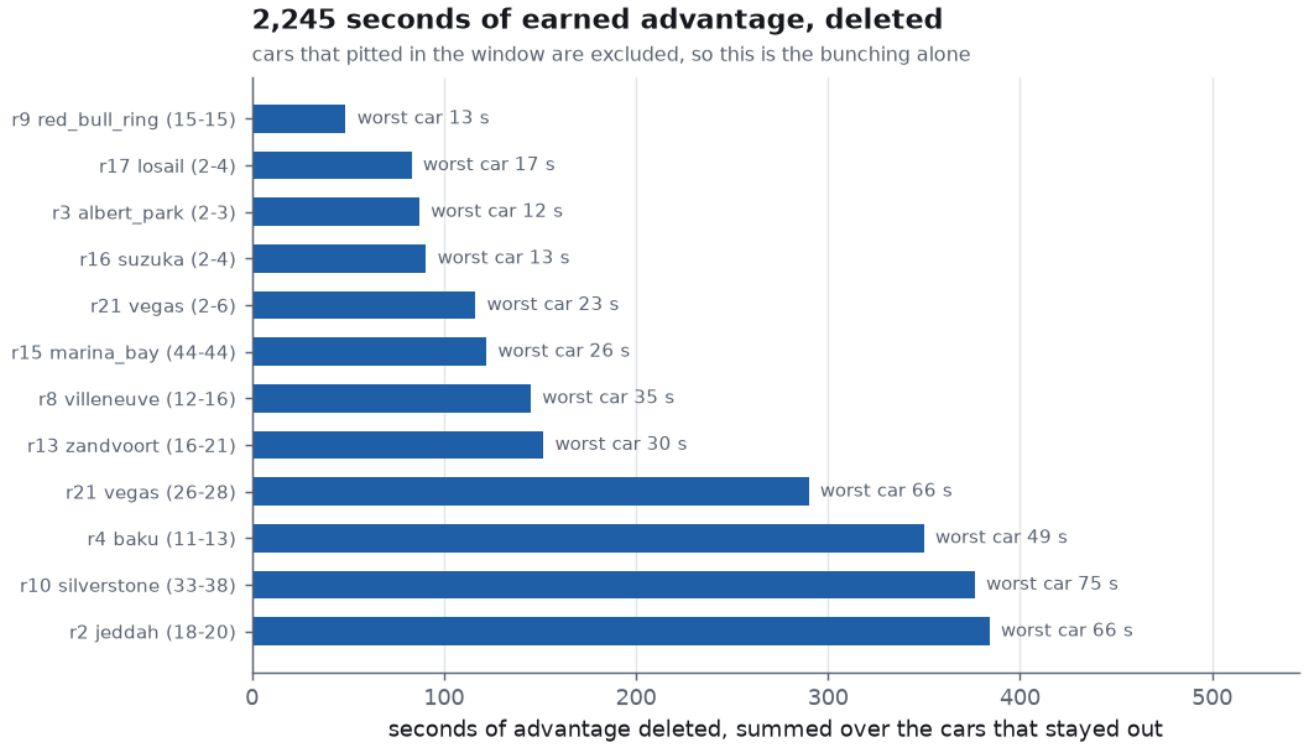
Which group a car was in depended on where it happened to be in its stint when somebody else crashed. That is the lottery in the title: not the discount, but who is eligible for it.



And what the bunching deleted

Counting only the cars that did not pit anywhere in the window, so this is the bunching alone and not the stops.

2,245 seconds of earned advantage deleted across the season.



race	laps	cars	seconds erased	worst hit	leader's lead
r2 jeddah	18-20	7	384	66	6.0 → 2.1
r3 albert_park	2-3	14	87	12	2.4 → 0.9
r4 baku	11-13	9	350	49	3.1 → 1.5
r8 villeneuve	12-16	7	145	35	4.0 → 1.2
r9 red_bull_ring	15-15	7	49	13	6.6 → 17.8
r10 silverstone	33-38	7	376	75	9.7 → 2.1
r13 zandvoort	16-21	13	152	30	3.3 → 1.2
r15 marina_bay	44-44	11	122	26	0.9 → 1.4
r16 suzuka	2-4	14	91	13	3.1 → 1.7
r17 losail	2-4	13	83	17	2.8 → 1.3
r21 vegas	2-6	14	116	23	1.7 → 1.8
r21 vegas	26-28	9	290	66	11.4 → 0.6

What this does not do

It does not know where the safety cars actually were — it infers them, and the three-way split leaves five periods unclear rather than forcing them. It cannot separate a full safety car from a virtual one, which have different slowdowns and different prizes. It prices a stop against staying out, which is not the same as the positions gained, because that depends on who else stopped. And it is one season: eighteen neutralisations is why the interval on the prize is as wide as it is.

step	stops
stops	912
priced	763
refused: no green line for the race	0
refused: no usable pace factor	8
refused: in-lap or out-lap not recorded	0
refused: the race was stopped on one of them	120
refused: nobody stayed out to measure the slowdown	21

Lap times, pit stops and results from the Jolpica mirror of the Ergast schema.